

**1st DRAFT**

# 2nd DRAFT

## I. THE PROBLEM DOMAIN

### 1.1 Statement of the Problem

Redeemed Route Logistics Inc. currently manages most of its operational and financial records manually through Microsoft Excel, paper documents, and other communication and storage channels. The company does not have a formal integrated information system for its operations and finance, making it difficult to quickly retrieve and organize important records.

This affects the way employees monitor deliveries, trucks, expenses, revenue, profitability, and Proof of Delivery (POD) documents. Important information may need to be searched from different sources before it can be used for reporting or decision-making.

The proposed RouteX system will address this problem by providing one centralized web-based platform where authorized employees can record, search, update, and retrieve operational and financial information.

### 1.2 Background and Rationale

Redeemed Route Logistics Inc. provides trucking and freight transportation services, including truck booking, delivery scheduling, route coordination, and dispatching.

Based on the team's interviews and analysis, the company has established operational procedures and personnel but still relies mainly on manual record management. This creates difficulties in maintaining organized records, monitoring delivery information, and analyzing financial performance.

RouteX is proposed as a partially automated system. Users will still enter and update information manually, while the system will automatically handle repetitive tasks such as calculating delivery margins, organizing records, searching information, and generating reports.

The system will therefore focus on improving the company's existing workflow rather than completely replacing human decision-making.

### 1.3 Objectives

#### General Objective

To develop a partially automated web-based system that centralizes the company's operational and financial records and improves the retrieval, management, and analysis of delivery-related information.

## Specific Objective

RouteX will include:

1. Centralize delivery, truck, driver, client, and financial records.
2. Allow authorized users to create, update, search, and retrieve records using relevant filters such as delivery ID, plate number, driver, client, and date.
3. Automatically calculate delivery profitability based on recorded revenue and expenses.
4. Allow users to upload and manage Proof of Delivery (POD) documents.
5. Record and manage company expenses not directly associated with a specific delivery.
6. Generate delivery, revenue, expense, and profitability reports.
7. Implement authentication, role-based access control, and activity logs for security and accountability.

## 1.4 Significance and Scope

### Significance

For the organization, RouteX provides a centralized source of operational information and supports better access to business data.

- **Management** can gain clearer visibility into revenue, expenses, and profitability.
- **Finance** can organize financial records and generate reports more efficiently.
- **Operations** can manage deliveries, trucks, and POD documentation in one system.
- **Customer Service** can quickly retrieve delivery, truck, and client information.
- **HR/Administration** can manage users and system access.

### Scope

The system will be developed as an **online web application** accessible through modern web browsers. It will cover:

- **Delivery Management** – Record and manage delivery details, including truck, driver, client, route, date, status, revenue, expenses, and POD, with automatic delivery margin calculation.
- **Transactions Management** – Record and manage company expenses not directly related to a specific delivery.
- **Search and Record Retrieval** – Search and retrieve delivery and transaction records using relevant information such as delivery ID, plate number, driver, client, and date.
- **Reports and Profitability Analysis** – Generate delivery, revenue, expense, and profitability reports with printable and export functions.
- **Security and Administration** – Provide login authentication, role-based access, user permissions, audit trails, and activity logs.



### III. REVIEW OF EXISTING ALTERNATIVE

| Features                         | CasinFreight | Routix TMS | Nexii Logistics Management System | HashMicro Transportation Management | SwiftDash Deliveries | RouteX |
|----------------------------------|--------------|------------|-----------------------------------|-------------------------------------|----------------------|--------|
| Delivery Management              | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | ✓      |
| Delivery / Trip Records          | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | ✓      |
| Truck / Vehicle Records          | ✓            | ✓          | —                                 | ✓                                   | ✓                    | ✓      |
| Driver Records                   | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | ✓      |
| Client / Customer Records        | ✓            | ✓          | —                                 | ✓                                   | ✓                    | ✓      |
| Dispatch Management              | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | —      |
| Route / Trip Planning            | ✓            | ✓          | —                                 | ✓                                   | ✓                    | —      |
| Delivery Status                  | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | ✓      |
| Proof of Delivery (POD)          | ✓            | ✓          | ✓                                 |                                     | ✓                    | ✓      |
| Revenue / Delivery Fee Recording | ✓            | ✓          | —                                 | ✓                                   | ✓                    | ✓      |
| Delivery / Trip Expenses         | —            | ✓          | —                                 | ✓                                   | —                    | ✓      |
| Company / General Expenses       | —            | —          | —                                 | ✓                                   | —                    | ✓      |
| Profitability / Margin Analysis  | —            | ✓          | —                                 | ✓                                   | ✓                    | ✓      |
| Search / Record Retrieval        | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | ✓      |
| Operational Reports              | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | ✓      |
| Financial / Cost Reports         | ✓            | ✓          | —                                 | ✓                                   | ✓                    | ✓      |
| User Authentication              | ✓            | ✓          | ✓                                 | ✓                                   | ✓                    | ✓      |
| Role-Based Access                | —            | ✓          | —                                 | ✓                                   | ✓                    | ✓      |
| Activity / Audit Logs            | —            | ✓          | —                                 | —                                   | ✓                    | ✓      |

|                  |   |   |   |   |   |   |
|------------------|---|---|---|---|---|---|
| Web-Based System | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
|------------------|---|---|---|---|---|---|

## IV. Approach to be taken in this Subject

### 4.1 Theoretical Framework / Method to be Used

The project will use a combination of the Iterative Model and Prototype Model for software development. The Prototype Model will be used to create an initial representation of RouteX that demonstrates its proposed features, interface, and workflow. The Iterative Model will then guide the continuous development, testing, and refinement of the system through repeated cycles. Each iteration will allow the system to be improved based on identified requirements and user feedback. For system design, process models such as flowcharts and data flow diagrams will be used to describe system processes, while an Entity-Relationship Diagram (ERD) will be used to represent the relationships among delivery, truck, driver, client, expense, and user records. This approach supports the development of RouteX as a partially automated system that centralizes operational and financial records while automating tasks such as calculations, record searching, and report generation.

### 4.2 Rationale for the Framework / Methods

- The combination of the **Prototype Model and Iterative Model** is appropriate for the development of RouteX because the system contains several interconnected functions that need to be presented, evaluated, and refined throughout the development process. The team used the Prototype Model to create an initial version of the proposed system that presents its main features, interface, and workflow. The prototype provides the client with a clear representation of how RouteX is expected to function and allows the proposed system to be reviewed before proceeding with further development.
- The **Iterative Model** is used as the main development approach because RouteX requires continuous improvement throughout the development process. The system will be developed through repeated cycles of planning, development, testing, evaluation, and refinement. Each iteration allows the team to identify issues, improve existing functions, and incorporate feedback or additional requirements. This approach is suitable because the system's requirements and user needs may become clearer as the system is reviewed and tested.
- The combined approach also supports the project's goal of improving the company's current manual processes. Redeemed Route Logistics Inc. currently relies on Microsoft Excel, paper documents, and other storage channels for managing operational and financial records. Through the use of a prototype and iterative development, RouteX can gradually be refined into a functional web-based system that centralizes delivery and financial records while partially automating tasks such as profitability calculations, record retrieval, and report generation.

### 4.3 Technologies you plan to consider or use

The table below presents the selected technologies and their specific roles in the development of RouteX.

| Layer | Technology | Why it fits RouteX |
|-------|------------|--------------------|
|-------|------------|--------------------|

| Category              | Technology                              | Purpose                                                                                                        |
|-----------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Web Framework         | Next.js                                 | Used to build the RouteX web application and handle frontend and server-side functionality.                    |
| UI Framework          | React                                   | Used to create reusable and interactive user interface components.                                             |
| Programming Language  | TypeScript                              | Used to provide type safety and improve code maintainability.                                                  |
| Styling Framework     | Tailwind CSS                            | Used for responsive and consistent user interface styling.                                                     |
| Component Library     | shadcn/ui                               | Used to provide reusable and customizable UI components.                                                       |
| Server-Side Functions | Next.js Actions & Server Route Handlers | Used to process business logic, form submissions, and server-side operations.                                  |
| Authentication        | Auth.js (NextAuth.js)                   | Used to manage user authentication and session handling.                                                       |
| Database              | PostgreSQL (Supabase)                   | Used to store and manage operational, financial, and administrative records.                                   |
| File Storage          | Supabase Storage                        | Used to store and manage uploaded documents, attachments, and supporting files utilized throughout the system. |
| Deployment Platform   | Vercel                                  | Used to host and deploy the application.                                                                       |

## VI. THE PROBLEM DOMAIN